Measurement resolution constrains fine-grained perturbation prediction

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Abstract

Virtual-cell models are increasingly evaluated at finer biological resolution, yet benchmarks rarely test whether measurements reproducibly resolve the scored distinctions. We introduce PertResolve, a framework that separates perturbation detection, identification and response prediction, and interprets model scores against the distinctions the measurement supports. Across 470 coding-variant conditions of TP53, KRAS, GATA1 and JAK1 (321,043 cells), feature-based predictors recovered transcriptional direction, as did a gene-mean baseline with no allele information, but neither they nor released perturbation models reliably identified alleles. Split-half references showed that similar scores had different causes: TP53 and KRAS were measurement-limited, whereas JAK1 carried reproducible allele structure exceeding the best feature-based predictor by 0.36 in perturbation discrimination score. Across 31 configurations from 14 public resources, detection depended on perturbation signal relative to within-condition variation, whereas identification depended on separation from nearby competitors. Measurement resolution therefore determines which distinctions perturbation prediction can evaluate and how benchmarks should be designed.

Virtual-cell models aim to predict how cellular states change in response to genetic, chemical and environmental perturbations, with the broader goal of simulating cellular behaviour across molecular and cellular contexts[1–3]. Current approaches include generative models[4,5], graph-based predictors[6], flow-matching methods[7] and pretrained single-cell models[8–10]. For genetic perturbations, however, widely used benchmarks remain largely gene-centred, encoding an intervention by the targeted gene[11–14]. This abstraction collapses protein-coding variants of the same gene even though such variants can differ in protein structure, biochemical activity, molecular interactions and downstream regulation[15–17]. A model can therefore recover a transcriptional programme shared across variants while capturing little of the response that distinguishes one variant from another. At variant-level resolution, evaluation thus depends not only on what a model predicts, but also on whether the experiment can reproducibly distinguish the variant-specific responses being compared.

Variant-resolved single-cell assays now make this problem experimentally accessible by measuring high-dimensional transcriptional responses to many coding variants in parallel[18–22]. Here, allele denotes

an experimentally assigned protein-coding variant condition rather than a phased genomic haplotype. Whereas multiplexed assays of variant effect and deep mutational scanning typically summarize each variant by a fitness, activity or molecular-function score[23–25], allele-resolved perturbation prediction seeks to predict the variant's high-dimensional cellular response. At this resolution, a high response-similarity score can reflect recovery of a gene-shared programme without showing that the correct allele has been resolved. This motivates separating three questions: whether a perturbation can be reproducibly distinguished from its reference, whether it can be distinguished from sibling alleles, and whether the response of a held-out allele can be predicted from its molecular properties.

This separation exposes a prerequisite that perturbation benchmarks usually leave implicit: the experimental measurement must itself reproducibly support the biological distinction that the benchmark is intended to evaluate. Measured response profiles are finite-sample estimates whose reproducibility depends on experimental design and within-condition variation[26–28]. If the requested distinction cannot be reproduced across disjoint cell samples, evaluation at that resolution is measurement-limited. If the distinction is reproducibly present in the data but remains unrecovered by the predictor, the setting is model-limited. Measurement resolution therefore determines the level of biological specificity at which model scores can be meaningfully interpreted. Coding variants provide a stringent setting in which to expose this problem, but the same principle applies whenever perturbation benchmarks are asked to distinguish closely related perturbations or cellular states.

Here we introduce PertResolve, a measurement-resolution framework for fine-grained perturbation prediction, together with PertResolve-Bench and PertResolve-Eval. PertResolve-Bench combines an allele-resolved benchmark of 470 protein-coding variant conditions in TP53, KRAS, GATA1 and JAK1 with a broader panel of 31 perturbation configurations from 14 public resources. In the allele-resolved setting, models recovered transcriptional direction but identified alleles at near-chance rates, and these similar scores arose for different reasons: TP53 and KRAS measurements were indistinguishable from chance on a split-half reference, whereas JAK1 measurements reproducibly separated 78.5% of sibling pairs and exceeded the best feature-based predictor by 0.36 in perturbation discrimination score (PDS). Across public resources, detection depends on perturbation signal relative to within-condition variation, whereas identification depends on separation from nearby competitors. Finally, we show how pilot measurements, candidate composition, sampling depth and benchmark size inform different aspects of perturbation-benchmark design.

Results

Allele-level model evaluation requires reproducibly resolved measurements

Detection and identification characterize the resolution of the measurement, whereas prediction of a held-out allele is the model-level problem (Fig. 1a). Model performance is interpreted only against distinctions that the measurement reproducibly supports.

The allele-resolved arm of PertResolve-Bench comprises 470 protein-coding variant conditions across TP53, KRAS, GATA1 and JAK1, together with their dataset-specific wild-type or control cells, totalling 321,043 cells profiled by Perturb-seq, base editing and scSNV-seq (Fig. 1b,c)[18–20]. The four datasets differ jointly in gene, assay and cellular context, and we therefore treat them as four measurement regimes rather than as four genes.

The prediction task holds out a coding variant from model training and asks the model to predict its population-average transcriptional response from molecular properties alone (Fig. 1d,e). Variants of the same gene remain distinct targets: TP53 R175H, R273C and R248W, for example, are predicted separately. For variant v, the target δ_v is the pseudobulk expression difference between variant and dataset-specific reference cells, and the model predicts $\hat{\delta}_v$ from biophysical or protein-language-model features without using the held-out variant's response cells (Methods).

Predictions are evaluated on complementary axes of response recovery and allele identification (Fig. 1f), under the generalization settings in Fig. 1g. Pearson-δ measures agreement with the observed transcriptional direction. The perturbation discrimination score (PDS) ranks the true measured response among all same-gene candidates and therefore tests full-pool allele identification; chance is 0.50. Differential-expression fidelity provides secondary biological readouts. We first asked whether predictors recover allele-specific information beyond the gene-shared perturbation programme.

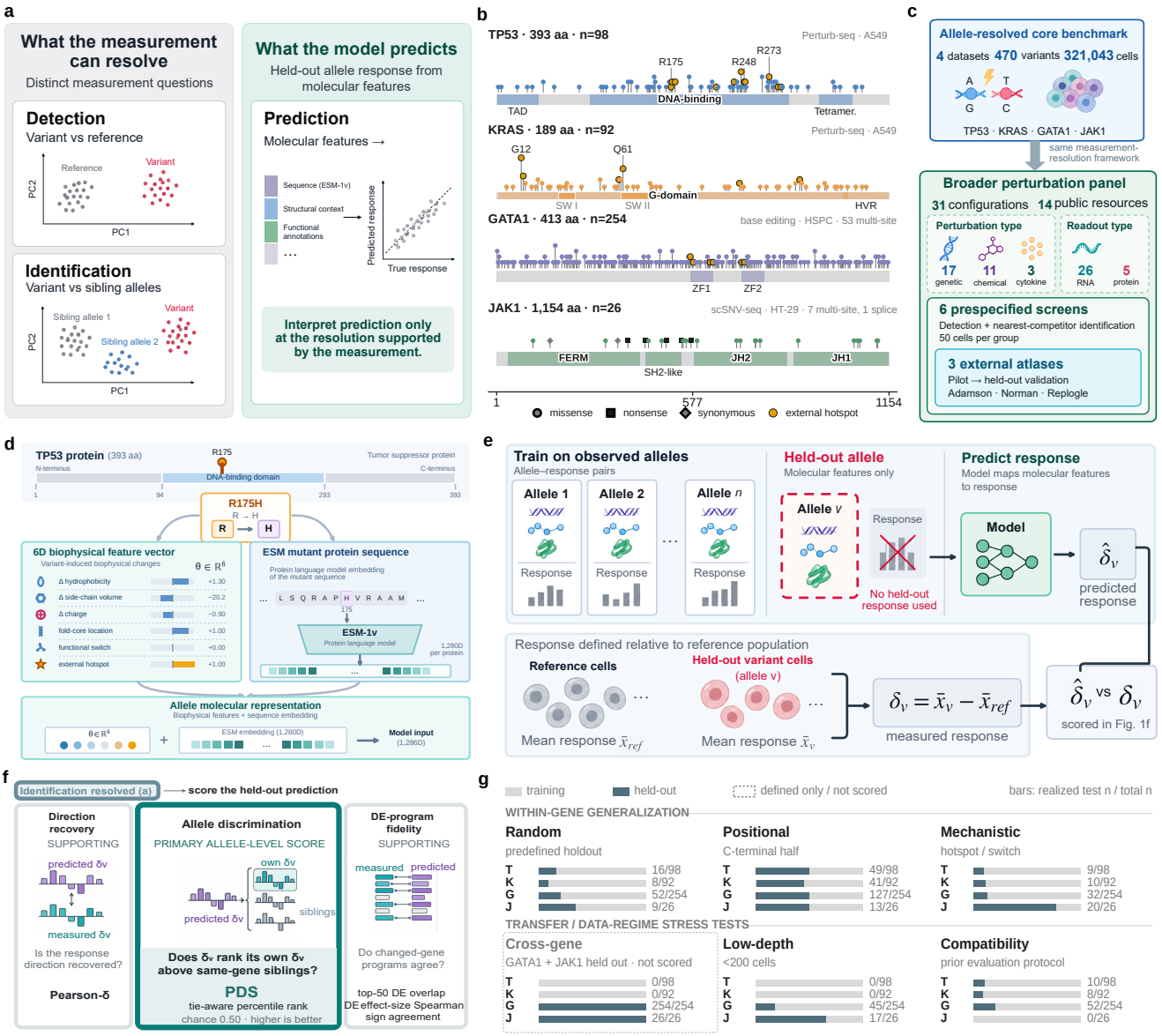


Figure 1: **PertResolve separates measurement resolution from fine-grained perturbation prediction. a**, PertResolve separates perturbation detection relative to the reference, identification among competing perturbations and prediction of a held-out perturbation. **b**, The allele-resolved arm of PertResolve-Bench comprises 470 protein-coding variant conditions across TP53, KRAS, GATA1 and JAK1 profiled by Perturb-seq, base editing and scSNV-seq. Marker shape denotes variant consequence; amber marks external hotspot annotations. **c**, PertResolve-Bench comprises two complementary analysis arms: an allele-resolved benchmark of four datasets, 470 variants and 321,043 cells, and a broader perturbation panel of 31 configurations from 14 public resources, including six prespecified screens and the three gene-perturbation atlases used for the pilot analysis. **d**, Molecular representations of a held-out allele: a six-dimensional biophysical vector θ and protein-language-model embeddings derived from the mutant sequence. **e**, Prediction task. Variant and reference cells define the measured response δ_v; the model predicts $\hat{\delta}_v$ from molecular features without access to the held-out variant's response cells. **f**, PertResolve-Eval reports transcriptional direction recovery (Pearson-δ), full-pool allele identification (PDS) and differential-expression fidelity. PDS, the primary allele-level score, ranks the true measured response among same-gene candidates; the other two are supporting axes. **g**, Evaluation settings, grouped as within-gene generalization and transfer or data-regime stress tests. Four primary scored settings test random holdout, positional extrapolation, mechanistic extrapolation and low-depth evaluation; compatibility reproduces a prior protocol, and cross-gene transfer is not scored (Methods). Bars show held-out fractions and counts.

Perturbation direction is recovered while full-pool allele identification remains near chance

Scoring each held-out prediction on both axes separated direction recovery from allele identification (Fig. 2a). We evaluated 18 feature-based predictors spanning six model heads and three feature spaces, together with the Gene-mean baseline and the WT-null reference (Fig. 2b,c). Across the 19 non-null predictors, Pearson-δ ranged from 0.58 to 0.67, whereas PDS ranged from 0.47 to 0.49; no predictor showed reliable above-chance full-pool allele identification.

Response decomposition explained the divergence. Within each cell partition, we separated the measured profiles into their mean across same-gene variants and the deviations from that mean. Averaged across partitions, the allele residual contributed 28%, 41%, 71% and 49% of the mean squared response norm in TP53, KRAS, GATA1 and JAK1, respectively (Fig. 2d; Methods). These shares include cell-sampling variation, which the split-half references below separate from reproducible allele signal. After measured and predicted profiles were centred separately within each evaluated gene cohort, residual Pearson-δ was near zero across predictors (-0.08 to 0.03). The Gene-mean baseline, which contains no allele-specific molecular information, matched the direction recovery of the best feature-based predictor (0.668 versus 0.673). Residual PDS, computed separately using a training-only gene mean, likewise remained close to its calibrated null (Fig. 2e). The direction recovered by the evaluated predictors was therefore dominated by the transcriptional programme shared across variants of the same gene.

Differential-expression fidelity showed the same loss of information with increasing biological specificity (Fig. 2f). Within the per-variant sets of 50 genes with the largest absolute Welch t-statistics, predicted and observed effects had the same sign for 68% of genes on average. Agreement was weaker for effect-size ordering (DE effect-size Spearman $\rho = 0.27$) and for the identity of responding genes (top-50 DE overlap, 15%).

Full-pool allele identification remains weak across variant representations and model classes

The dissociation between direction recovery and full-pool identification also persisted across richer molecular representations. Protein-language-model embeddings, mutation-aware ESM2 representations and AlphaMissense features preserved direction recovery without producing reliable full-pool allele identification (Supplementary Note 2)[29,30].

Released perturbation-prediction frameworks evaluated on the same held-out variants with the same allele-level scoring showed the same limit (Fig. 2g; Supplementary Table 3). Biolord and CellFlow, together with the allele-conditioned adaptation of scGen, yielded PDS values of 0.47–0.49 with confidence intervals overlapping 0.50. PerturbNet reached a PDS of 0.530 (95% confidence interval, 0.504–0.555), above the uniform-matching expectation of 0.50 but not above its task-matched permutation null (null mean 0.515; one-sided $P = 0.070$; Statistical analysis). scVIDR, whose GATA1 scaling uses the held-out variant's measured response, is shown as a response-informed control (Supplementary Note 3). Gene-keyed interfaces such as scGPT, GEARS and STATE build in the gene-centred encoding of current benchmarks: they condition on perturbed-gene identity, so the 470 variant conditions collapse onto four conditioning keys and sibling alleles receive the same prediction (Supplementary Note 3).

A more operational task, placing the correct allele in a shortlist of k candidates, yielded the same conclusion (Fig. 2h). No method exceeded its query-matched chance expectation at $k = 1, 3, 5$ or 10, and 54 of the 100 method–k combinations fell below it. A near-chance model score alone, however, cannot distinguish failure to learn reproducible allele structure from a measurement that does not resolve that structure. We therefore quantified the allele-level resolution of the measurements themselves.

Measurement resolution separates measurement-limited from model-limited allele prediction

To place measurement reproducibility and model performance on the same allele-identification scale, we constructed a split-half reproducibility reference: each held-out variant's cells were divided into disjoint groups, and one half was scored against the other with the same full-pool PDS evaluation used for the prediction models (Fig. 3a,b).

The split-half reference revealed sharply different measurement regimes. TP53 and KRAS remained indistinguishable from chance, so these measurements did not reproducibly support full-pool allele identification at the evaluated depth. GATA1 showed modest reproducible structure, whereas JAK1 showed strong allele-level reproducibility; the measurement–model gap was 0.069 PDS in GATA1 and

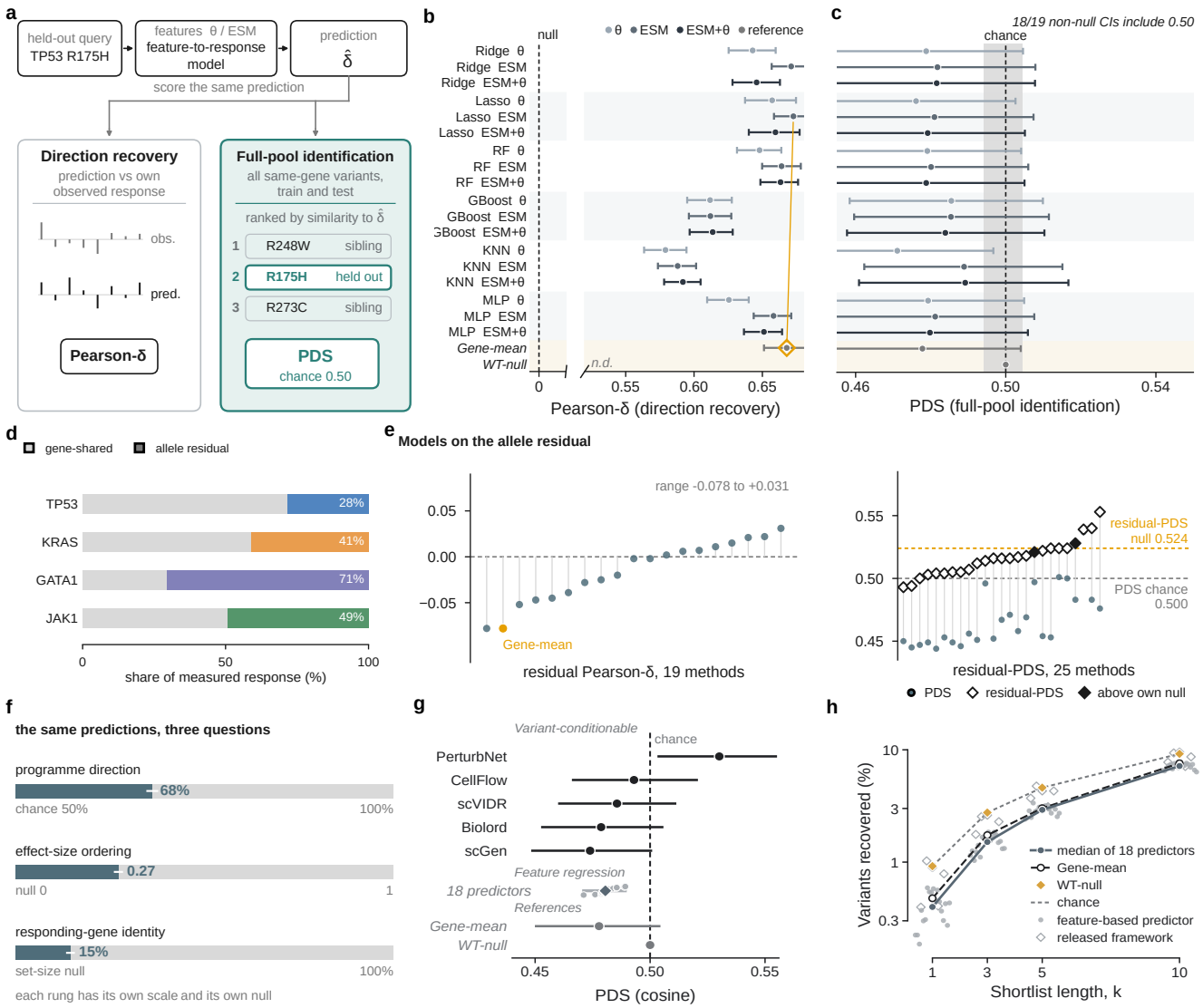


Figure 2: **Evaluated predictors recover gene-shared response while full-pool allele identification remains near chance.** **a**, Complementary evaluation by transcriptional direction recovery (Pearson-δ) and full-pool allele identification (PDS; chance, 0.50). **b**, Pearson-δ for 18 feature-based predictors and two references. Points, means; whiskers, 95% bootstrap confidence intervals over distinct held-out variants ($n = 308$, or 305 for sequence-dependent predictors). Gene-mean is highlighted; n.d., undefined for WT-null. **c**, PDS for the same predictors; intervals as in **b**. Grey band, range of held-out-only permutation means (0.494–0.505; Methods). **d**, Gene-shared and allele-residual shares of the mean squared measured-response norm, averaged over 15 cell partitions. Sample sizes are 98 TP53, 92 KRAS, 254 GATA1 and 26 JAK1 variants (Methods). **e**, Left, descriptive residual Pearson-δ after separate cohort centring. Right, full-response PDS (circles) and training-centred residual PDS (diamonds). Filled diamonds, unadjusted permutation $P < 0.05$; multiplicity adjustment and permutation resolution are described in Methods. Lines mark the respective nulls. **f**, Differential-expression fidelity: sign agreement, DE effect-size Spearman correlation and top-50 DE overlap. Each metric uses its own scale and null. Each predictor's value is its mean over held-out evaluations pooled across datasets and settings; bars, means across 18 predictors; white ticks, ranges. **g**, Multi-seed PDS for five released-framework implementations; intervals bootstrap each method's distinct-variant query cohort, with coverage reported in Supplementary Table 3. Lower points, feature-based predictors; diamond, their median. The PerturbNet estimate aggregates five evaluation settings over 303 distinct variants; its task-matched restricted-label permutation test gives $P = 0.070$. scVIDR is a response-informed control (Methods). **h**, Top-k allele shortlisting. Filled circles, feature-based predictors; open diamonds, released-framework implementations; line, feature-based median; references shown separately. scVIDR as in **g**. Dashed line, mean query-specific chance $\min(k/N_v, 1)$ over the 308 queries, with N_v the candidate-pool size; inference uses each method's own query-matched null (Methods).

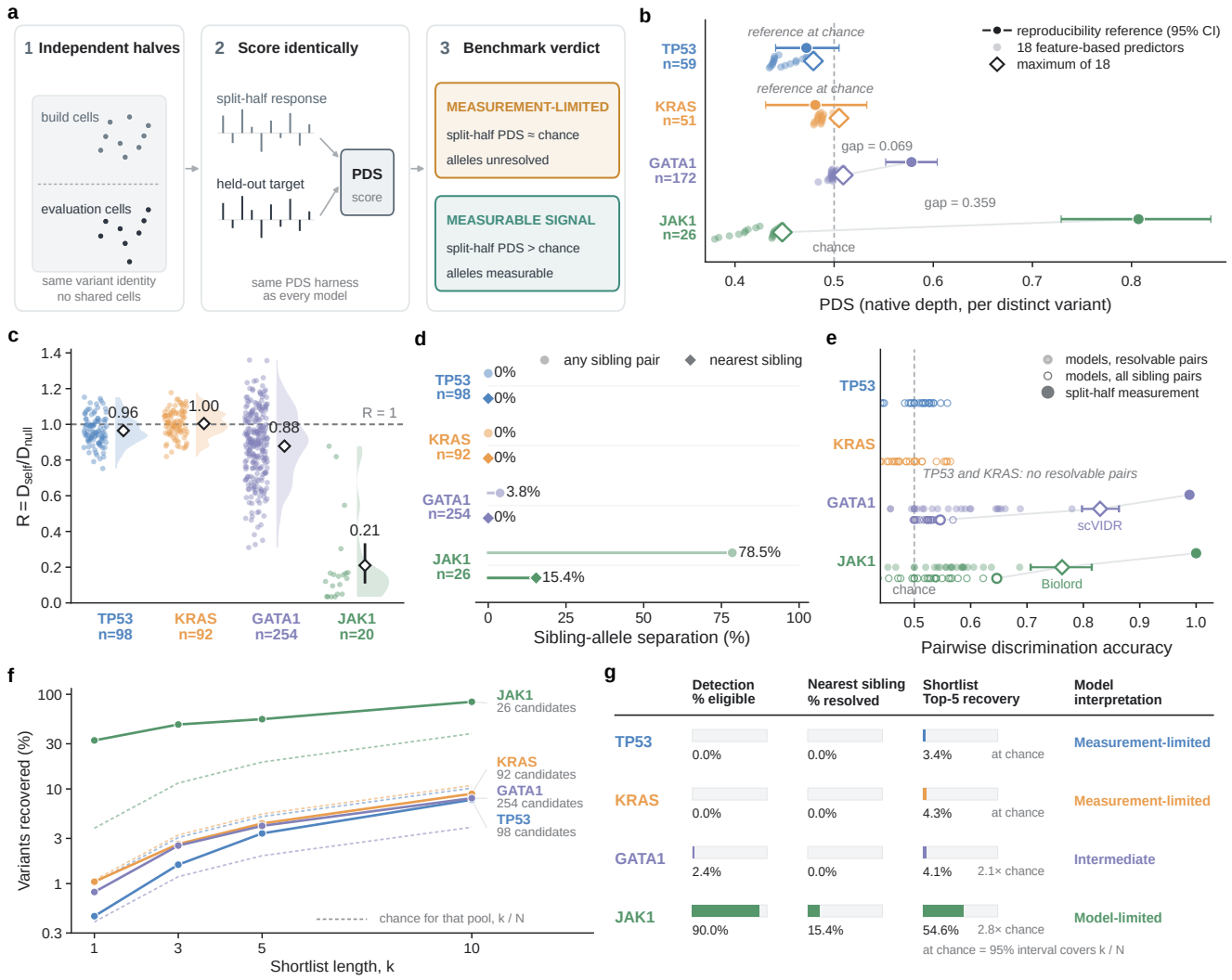


Figure 3: **Split-half measurement resolution distinguishes measurement-limited from model-limited allele prediction. a**, Split-half reproducibility reference: disjoint cell groups supply the query and target responses for the same full-pool PDS evaluation used for models. **b**, Reference and 18 feature-based predictors at native depth. Reference whiskers, 95% bootstrap confidence intervals over distinct held-out variants (n at left); open diamond, best predictor; dashed line, chance PDS (0.50). **c**, Detection-window ratio $R = D_{\text{self}}/D_{\text{null}}$, relating within-variant variation to variant-to-reference separation. Diamonds, dataset means; whiskers, 95% bootstrap confidence intervals; n, variants evaluable at native depth; $R = 1$, equal variation and signal. **d**, Fractions of sibling pairs separated beyond within-variant variation (circles) and variants resolving their nearest sibling (diamonds). **e**, Pairwise discrimination accuracy. Small filled markers show models on pairs independently identified as measurement-resolvable; small open markers show models on all scored pairs. The large filled circle shows the split-half measurement on resolved pairs. The labelled diamond marks the highest displayed implementation on resolved pairs, joined to its all-pairs value; the GATA1 diamond is the response-informed scVIDR control. Whiskers, 95% percentile-bootstrap intervals over 425 GATA1 and 116 JAK1 resolved pairs; variant-level intervals are given in Supplementary Note 2. Chance, 0.50. TP53 and KRAS have no resolved pairs and show only all-pairs scores. **f**, Split-half top-k recovery: fraction of held-out variants whose true allele ranks among the top k same-gene candidates. Dashed curves, chance k/N for each dataset (N = 26–254 candidates); logarithmic axis. **g**, Interpretation across identification tasks. *At chance* denotes a 95% recovery interval covering k/N.

0.359 in JAK1. The detection-window ratio $R = D_{\text{self}}/D_{\text{null}}$, the split-half within-variant distance divided by the variant-to-reference distance, lay near 1, where perturbation signal is comparable to within-condition sampling variation, for TP53 and KRAS, was intermediate for GATA1 and was substantially lower for JAK1 (Fig. 3c). No sibling pairs were reproducibly separated in TP53 or KRAS, compared with 3.8% in GATA1 and 78.5% in JAK1, and 15.4% of JAK1 variants also resolved their nearest sibling (Fig. 3d; Supplementary Table 4).

Among measurement-resolvable sibling pairs, the evaluated response-blind models recovered substantial allele-specific information (Fig. 3e). The split-half measurement discriminated these pairs with accuracies of 0.988 in GATA1 and 1.000 in JAK1, whereas the highest response-blind point estimates were 0.780 for Ridge-ESM in GATA1 and 0.762 for Biolord in JAK1. Their paired gaps to the measurement remained negative under a variant-node bootstrap that accounts for pairs sharing a variant (Supplementary Note 2). Thus, response-blind models recover allele-specific structure where the measurement supports the comparison, without reaching the measurement reference.

Between pairwise discrimination and full-pool identification lies the practical question of whether the measurement can place the correct allele in a useful shortlist (Fig. 3f). At $k = 10$, the split-half reference recovered 83.3% of JAK1 variants compared with a 38.5% chance rate, whereas TP53 and KRAS remained near chance. GATA1 illustrates the distinction between local and many-way resolution: although its small subset of resolvable sibling pairs was discriminated almost perfectly, the correct allele appeared in the top five for 4.1% of variants, compared with 2.0% by chance. High discrimination on selected pairs therefore does not imply useful identification across the full candidate pool.

Similar near-chance model PDS values therefore arise in fundamentally different regimes (Fig. 3b,g). TP53 and KRAS are measurement-limited at allele resolution; JAK1 is model-limited, because reproducible allele structure substantially exceeds model performance; GATA1 is intermediate, with a small resolvable subset but weak full-pool resolution. In JAK1, most of this structure lies within the span of the training responses: projecting each held-out response onto the measured training-variant responses gave a PDS of 0.690, against 0.482 for the best feature-based predictor and 0.792 for the split-half reference over the same four cell splits (Supplementary Note 2). Direction recovery orders the datasets almost oppositely: mean Pearson-δ of the 18 feature-based predictors decreases from 0.76 in TP53 to 0.38 in JAK1, while the fraction of reproducibly resolved sibling pairs increases from 0% to 78.5% and their mean PDS stays between 0.42 and 0.50 in every dataset. The dataset on which predictions appear strongest by direction recovery is the one that provides the least allele-level measurement resolution.

Measurement resolvability depends on perturbation signal relative to within-condition variation

We next asked what determines these measurement regimes. Detection depends on the separation between a perturbation and its reference relative to the variation obtained by repeatedly measuring the perturbation itself, which R summarizes (Fig. 4a); representative TP53 and JAK1 variants illustrate closed and open measurement windows (Fig. 4b).

Across the four allele-resolved datasets, nominal cell count did not order measurement resolution. Larger pseudobulk responses were associated with lower detection-window ratios (Fig. 4c). TP53 and KRAS had median depths of 929 and 1,000 cells per variant yet no detection-eligible variants, whereas JAK1 had a median of 104 cells per variant and 90% of its 20 variants evaluable at native depth were detection-eligible; GATA1 was intermediate at 2.4% (Fig. 4d). Subsampling 30–150 cells per half with perturbation identity held fixed preserved these regimes: TP53 and KRAS remained unresolved, GATA1 weakly resolved, and JAK1 was strongly resolved at the shallowest tested depth. Sampling depth over this range therefore did not explain the between-dataset differences in measurement resolution.

Allele identification depends instead on separation from nearby competing perturbations. Cross-fitted nearest-competitor separation relative to within-condition sampling variation followed the same ordering (Fig. 4e). TP53 and KRAS showed no reproducibly positive nearest-competitor separation at 25, 50 or 100 cells per group, whereas JAK1 remained positive at all three depths; GATA1 lay near the boundary. The datasets that differed in detection therefore also differed in the local response geometry required for allele identification.

The relative measurement regimes were preserved across five distance families and alternative detection criteria (Fig. 4f; Supplementary Fig. 1; Supplementary Note 1). Because detection and nearest-

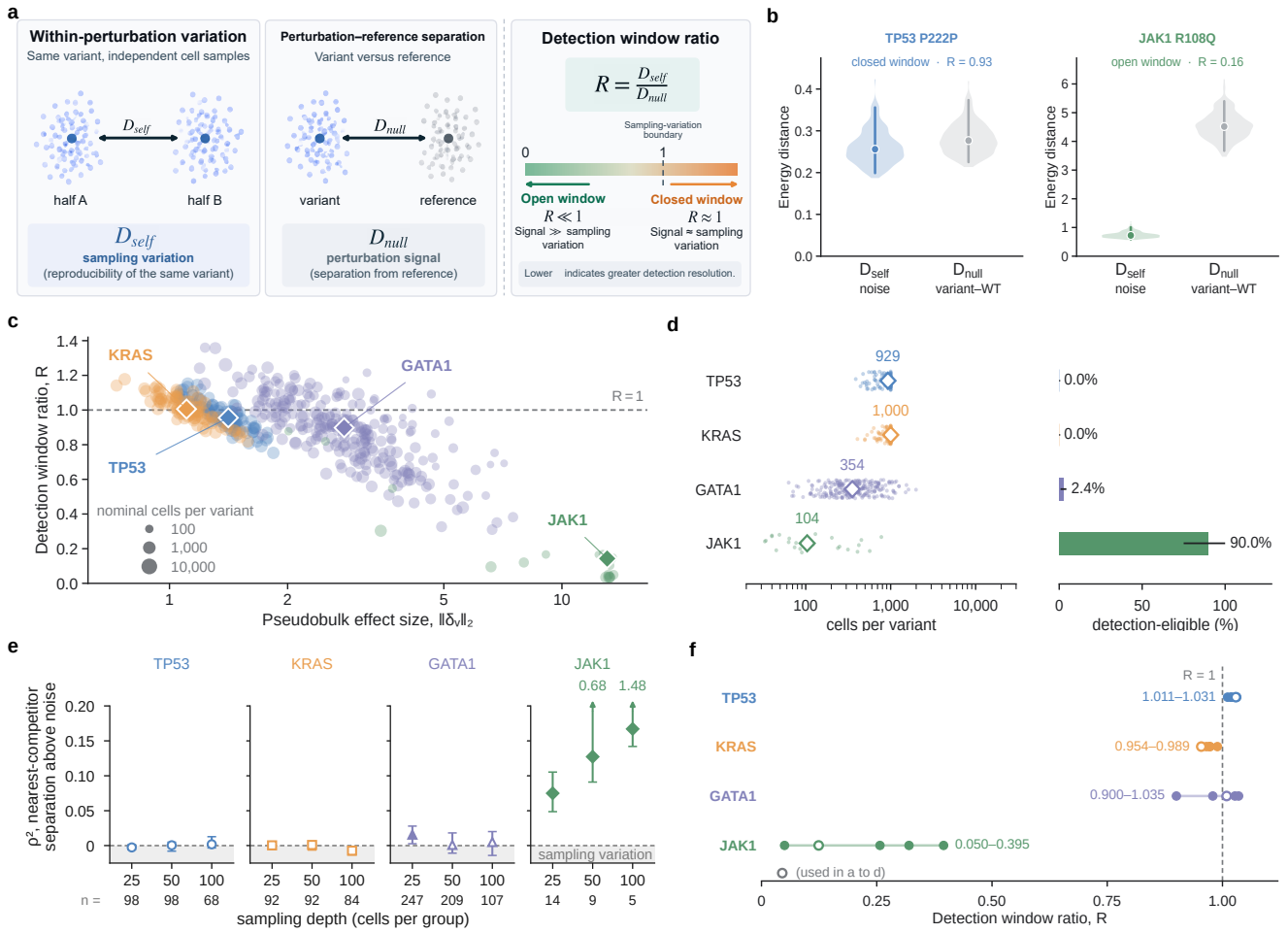


Figure 4: **Measurement resolvability depends on perturbation signal relative to within-condition variation.** **a**, Detection-window ratio $R = D_{\text{self}}/D_{\text{null}}$ (Fig. 3c); detection eligibility uses the $S > W$ criterion (Methods). **b**, Representative TP53 P222P and JAK1 R108Q windows. Distributions show distances across 50 split-half draws; y-axis scales differ. **c**, Detection-window ratio against pseudobulk response magnitude $\|\delta_v\|_2$. Marker area, nominal cell count; diamonds, dataset medians. **d**, Available cells per variant and detection-eligible fractions at native depth. Diamonds, median cell counts; bars and whiskers, fractions and 95% Wilson confidence intervals. Denominators are variants evaluable at native depth (JAK1, 20 of 26; Methods). **e**, Cross-fitted nearest-competitor separation relative to within-condition sampling variation at 25, 50 and 100 cells per group. Points, medians across variants; whiskers, 95% bootstrap confidence intervals; filled symbols, intervals entirely above zero. Zero, nominal no-signal reference (Methods). Lower numbers, evaluable variants; upper JAK1 numbers, confidence limits beyond the axis. Cohorts differ across depths because deeper groups require more cells. **f**, Detection-window ratios at 50 cells per half under five distance families: energy distance, maximum mean discrepancy, sliced Wasserstein, cosine and Euclidean distance. Rows, datasets; printed intervals, ranges across families; open symbols, energy distance used in **a**–**d** (Supplementary Note 1).

competitor resolution are defined from perturbation responses and their references rather than allele-specific annotations, PertResolve applies beyond coding variants. We therefore evaluated the broader perturbation arm of PertResolve-Bench across public genetic, chemical and cytokine perturbation resources.

Measurement resolution varies widely across public perturbation screens

We applied the same detection criterion to 31 perturbation configurations nested within 14 public resources, spanning genetic, chemical and cytokine perturbations and RNA and surface-protein readouts (Fig. 5a; Methods). The fraction of perturbations meeting the detection criterion spanned the full range from 0 to 1 (Fig. 5b), placing the allele-resolved datasets within a broad continuum of perturbation measurement resolution.

Among the 25 configurations with recorded cell counts, detection was not ordered by cells per perturbation: the deepest configuration (13,080 cells per perturbation) had a detection fraction of 0.256, whereas one with 1,587 cells reached 1.000 (Fig. 5c). Matched RNA–protein comparisons showed no consistent ordering by readout modality, with protein resolving more perturbations in two comparisons

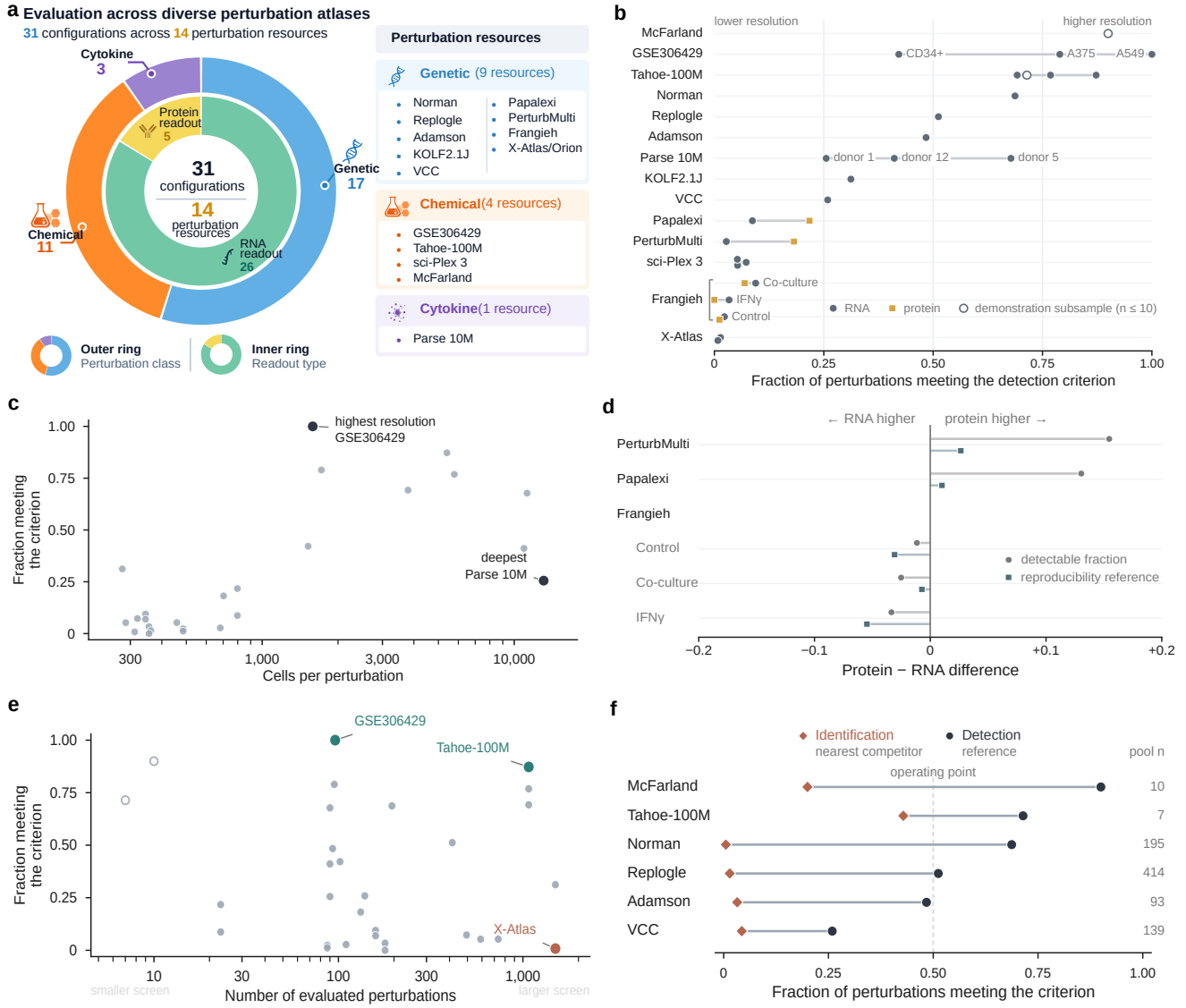


Figure 5: **Public perturbation screens span broad measurement-resolution regimes.** **a**, Analysis scope: 31 configurations nested within 14 resources, spanning genetic, chemical and cytokine perturbations and RNA and surface-protein readouts. **b**, Detection fractions grouped by resource. Rows are ordered by median; horizontal rules span within-row ranges. Frangieh 2021 contexts are shown separately. RNA fractions use 50 principal components of unnormalized counts (Methods). **c**, Cells per perturbation against detection fraction for the 25 configurations with recorded cell counts. **d**, Protein-minus-RNA differences in detection fraction and split-half reproducibility reference for five matched context–resource pairs from three resources. The PerturbMulti reproducibility comparison uses different candidate pools for RNA and protein. **e**, Evaluated perturbation count against detection fraction for all 31 configurations; logarithmic count axis. Coloured symbols identify the highest-resolution configuration and the highest- and lowest-resolution configurations among the ten largest. Open symbols, demonstration subsamples. **f**, Detection and nearest-competitor identification in six prespecified screens at 50 cells per group. Circles, fractions separable from reference; diamonds, fractions separable from the nearest competitor; connectors, within-screen differences; dashed line, 0.50 operating point. Rows are ordered by detection; candidate-pool sizes at right. McFarland and Tahoe-100M are demonstration subsamples of at most ten perturbations. VCC, Virtual Cell Challenge.

and RNA in three (Fig. 5d). Nor did screen size: among the ten largest configurations (414–1,500 perturbations), the detection fraction ranged from 0.008 to 0.873, whereas one of 96 perturbations reached 1.000 (Fig. 5e). Neither axis of screen scale is a proxy for whether a screen can be evaluated.

Detection and identification separated even more strongly when both criteria were evaluated in the six prespecified external screens at 50 cells per group (Fig. 5f; Supplementary Table 5). In the four full screens, 26–69% of perturbations were separable from their reference but at most 4.3% from their nearest competitor; the two demonstration subsamples of at most ten perturbations reached 20% and 43% on correspondingly small candidate pools, and no screen exceeded the 0.50 operating point for identification. Identification is also depth-dependent: on a fixed cohort, increasing one GSE306429 configuration from 50 to 200 cells per group raised its identification fraction from 0.085 to 0.557. Across pub-

lic resources, measurement resolution was not ordered by cell number, screen size, perturbation class or readout modality, and detection of a perturbation relative to its reference did not imply resolution among nearby competitors. We next asked whether these quantities can be estimated early enough to guide benchmark design.

Measurement resolution informs benchmark design

Benchmark design involves three distinct decisions: which perturbations are likely to be detectably measured, which competitors define the identification problem, and how many perturbations are needed to distinguish models (Fig. 6a).

Pilot-scale measurements ranked perturbations by detectability (Fig. 6b). In Adamson, Norman and Replogle, pilot scores from 25 or 50 cells, averaged over ten pilot draws and compared with the detection label aggregated over the matching disjoint evaluation draws (Methods), ranked detectable above undetectable perturbations with areas under the receiver operating characteristic curve (AUROCs) of 0.85–0.99 for a learned effect-size score and 0.84–0.98 for a training-free signal-to-variation rule. Identification among close competitors depends instead on competitor geometry, examined next.

Candidate-set size and candidate composition had different effects on attainable identification (Fig. 6c). Uniformly subsampling same-gene competitors left the mean split-half PDS unchanged from $K = 5$ to the full candidate pool, as expected for a percentile-ranked score. Restricting evaluation to the nearest competitors sharply reduced attainable identification: JAK1 fell from 0.807 over the full pool to 0.445 against its four nearest competitors, and GATA1 from 0.578 to 0.495. Controlled response geometries confirmed this dependence on local geometry. Across 1,440 configurations in which mean signal was controlled while response geometry varied, together with 12 measured gene-by-depth conditions, the split-half reproducibility reference tracked cross-fitted nearest-competitor separation more closely than mean pairwise separation (Spearman 0.961 versus 0.767; Fig. 6d). Local competitor separation, rather than average perturbation separation or candidate count, is therefore the more informative measure of attainable identification.

Pilot measurements and local competitor geometry can also inform sampling-depth projections: a calibration linking nearest-competitor separation to the split-half reproducibility reference returned finite projections for JAK1 and GATA1 and none for TP53 or KRAS, matching their measurement regimes (Fig. 6e; Supplementary Note 5).

Sampling depth remained consequential when perturbation identity was held fixed. Subsampling Replogle, Norman and Adamson to 50 cells per half reduced the detection-eligible fraction by 2.8, 8.5 and 24.0 percentage points, respectively (Fig. 6f). A common cell budget therefore does not produce a common measurement resolution, because screens differ in how much additional depth improves perturbation-to-reference separation.

Benchmark size formed a separate design axis (Fig. 6g). Among predictor pairs whose design ordering was realized in the full cohort, the smallest tested benchmark achieving a 0.9 ordering-recovery probability ranged from ≤ 5 perturbations for well-resolved measurements with large model differences to 160 perturbations when measurement reproducibility was near chance and the realized PDS difference was at most 0.01. The number of perturbations required for model comparison therefore depends jointly on measurement reproducibility and the performance difference to be resolved.

Discussion

Fine-grained perturbation prediction can be interpreted only where the experiment reproducibly distinguishes the states being compared. In our allele-resolved benchmark, models recovered the transcriptional direction shared by variants of the same gene but identified the correct allele among all same-gene candidates at near-chance rates, for different reasons in different datasets. TP53 and KRAS measurements did not reproducibly separate the alleles at the evaluated resolution, whereas JAK1 measurements separated 78.5% of sibling pairs, structure that the evaluated models have yet to recover in full-pool identification. On reproducibly separated sibling pairs, the best response-blind models discriminated alleles with accuracies of 0.78 in GATA1 and 0.76 in JAK1, against a chance level of 0.50, so a weak full-pool score can coexist with allele-level signal that models already capture.

The same distinction extends beyond coding variants. Across 31 configurations from 14 public re-

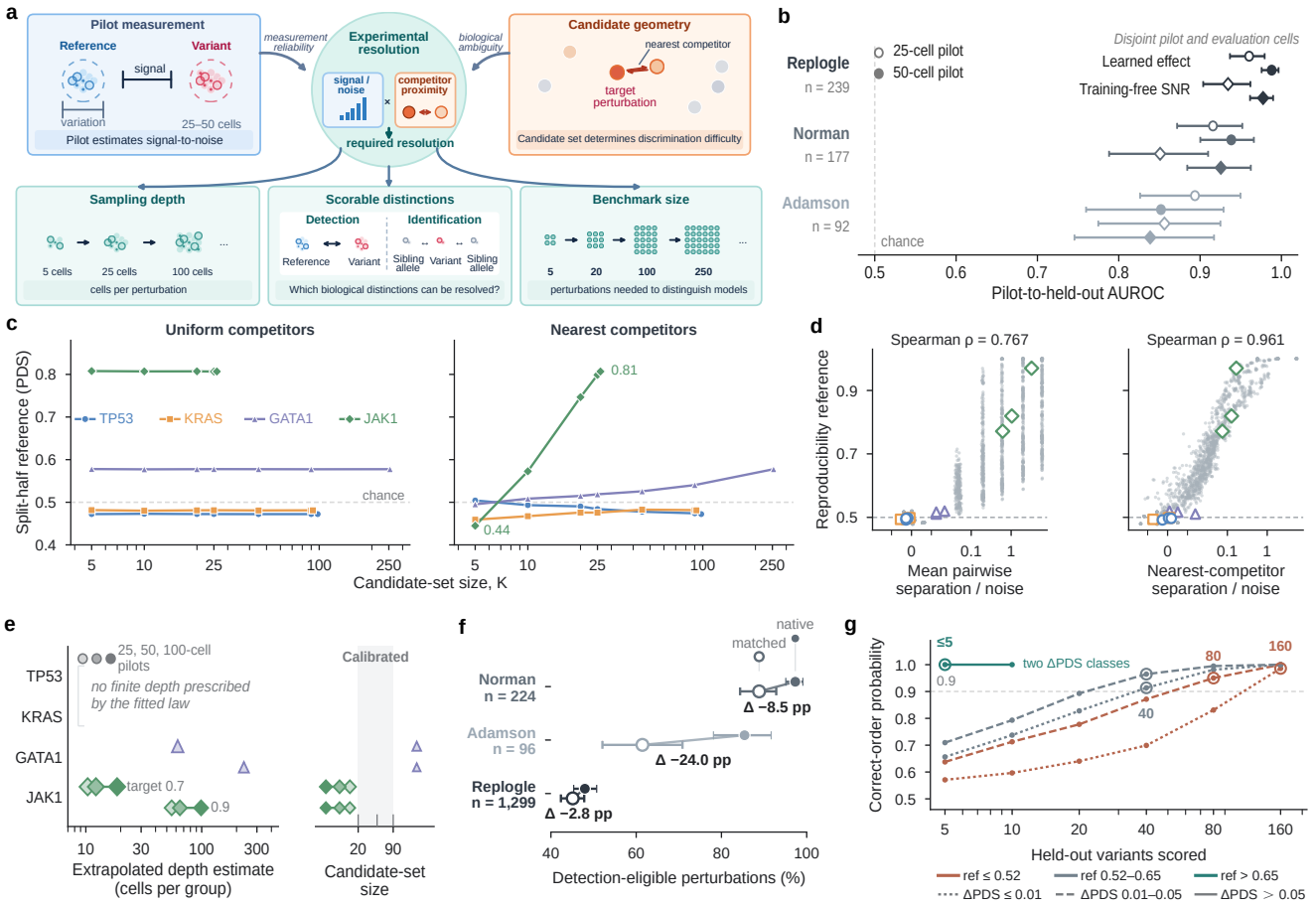


Figure 6: **Measurement resolution informs benchmark design.** **a**, Design framework linking pilot measurements, competitor geometry, sampling depth and benchmark size. **b**, AUROC of pilot scores (25 or 50 cells, averaged over ten pilot draws) against the detection label aggregated over ten matching, disjoint evaluation draws. Symbols distinguish learned effect-size and training-free signal-to-variation scores. Whiskers, 95% bootstrap confidence intervals over perturbations; dashed line, chance AUROC; n, evaluated perturbations. **c**, Split-half reproducibility reference under uniformly sampled competitors (left) and nearest-competitor restriction (right), varying candidate-set size K, including the target. **d**, Reproducibility reference against mean pairwise separation (left) and cross-fitted nearest-competitor separation (right). Grey points, 1,440 controlled response geometries; open symbols, 12 measured gene-by-depth conditions. Reported coefficients are Spearman correlations. **e**, Model-based depth projections from pilot nearest-competitor separation, with candidate-set size selecting the calibration stratum. Eight of 24 gene–target–pilot combinations return finite projections. The companion panel shows that all eight lie outside the calibrated candidate-set range (20–90). **f**, Native versus depth-matched detection at 50 cells per half in Norman, Adamson and Replogle. Whiskers, 95% paired-bootstrap confidence intervals; n, perturbations evaluated at both depths. **g**, Probability of recovering the controlled predictors' design ordering as benchmark size increases, grouped by measurement reproducibility and realized PDS difference. Recovery probabilities are summarized over gene–depth–predictor-pair conditions with a positive realized full-cohort PDS difference (Methods). Labels mark the smallest tested size reaching 0.9 recovery; ≤ 5 is an upper bound.

sources, detection of a perturbation relative to its reference depended on its response relative to within-condition variation, whereas identification also depended on how close the nearest competing perturbations were. Cell number, screen size and readout modality did not order detection, and a perturbation clearly separated from its reference was not necessarily separable from its nearest competitors. Resolution is therefore a property of a dataset together with the biological question, the readout, its representation and the comparison being evaluated.

These findings separate perturbation-benchmark design into three decisions. More cells sharpen each measured response but do not by themselves resolve biologically similar perturbations; the competitors included, more than their number, set the difficulty of identification; and benchmark size should follow the model difference that the study aims to resolve. Averaged pilot scores rank perturbations by likely detectability, whereas identification requires explicit assessment of nearby competitors. For model development, the most informative settings are those in which the measurement reproducibly supports a distinction that current models do not yet recover, as in JAK1. On a comparison the experiment does not

resolve, a low model score reflects the measurement and carries little information about model capacity.

These conclusions have a defined scope. The allele-resolved analysis includes four dataset–assay–context configurations in which gene identity, experimental technology and cellular context vary together, so the observed regimes are properties of these configurations rather than of TP53, KRAS, GATA1 or JAK1 as genes. The split-half analyses measure reproducibility across cell samples from the same experiment, and the split-half reference is an empirical reference that predictors which denoise measurements or borrow information across conditions can exceed. Measurement resolution is defined for the readout and representation used for evaluation, and biological effects that are weak in transcriptional space may be resolved by other molecular or phenotypic readouts. The pilot scores are averaged over repeated draws within each screen, and a prospective test with one fixed pilot remains to be done.

Before a model score is interpreted at fine biological resolution, the experiment must be shown to resolve that distinction. Reporting measurement resolution alongside model performance separates what the experiment cannot distinguish from what the model has not yet learned.

Methods

PertResolve-Bench: datasets and preprocessing

Allele-resolved benchmark. All analyses used publicly available single-cell perturbation datasets; no new experiments were performed. The primary unit of prediction in the allele-resolved benchmark was the protein-coding variant condition. The allele-resolved component included datasets with genotype-linked single-cell expression, multiple coding variants of the same gene sufficient for within-gene evaluation, and compatibility with a common held-out-variant protocol. TP53, KRAS, GATA1 and JAK1 met these criteria (Supplementary Table 6).

Data sources and eligibility. TP53 (98 variants) and KRAS (92 variants) were obtained from a Perturb-seq screen of coding variants in A549 lung adenocarcinoma cells (Gene Expression Omnibus GSE161824)[18]. GATA1 (254 variants) was obtained from a base-editing screen in human haematopoietic stem and progenitor cells[19]; we used the processed expression matrix distributed with PerturbNet[31] (Hugging Face repository `cyclopeta/PerturbNet_reproduce`). JAK1 (26 variants) was obtained from an scSNV-seq experiment in HT-29 colorectal cancer cells stimulated with interferon-gamma (European Nucleotide Archive PRJEB48915; processed objects from Zenodo record 10418435)[20]. The four datasets comprise 470 protein-coding variant conditions and 321,043 cells, including the dataset-specific wild-type or control cells retained after preprocessing.

Expression preprocessing. For TP53 and KRAS, processed count matrices, cell barcodes and variant-to-cell assignment tables were loaded from GSE161824, yielding a shared 939-gene expression space for the two A549 datasets. For GATA1, we used the PerturbNet-distributed AnnData object containing log-normalized expression over 2,477 highly variable genes. For JAK1, the released SingleCellExperiment object was loaded in R (Bioconductor). Cells were restricted to homozygous genotyped calls with unambiguous coding consequences (missense, splice-site, stop-gained or synonymous). The 2,000 most highly variable genes were selected using `scran::modelGeneVar` followed by `getTopHVGs`, and log-normalized counts were exported for downstream analysis.

Each dataset was analysed in its native expression space; expression genes were therefore not required to match across datasets. Within each gene and generalization split, expression features were standardized using only cells assigned to training variants and to the corresponding reference population, excluding all held-out-variant cells. The fitted transformation was then applied unchanged to training and held-out cells. The dataset-specific reference population, wild-type or control cells, served as the common reference for constructing perturbation-response profiles, and is denoted the reference throughout.

Variant representations. Each variant v was represented by molecular features rather than by a variant identifier, allowing prediction for variants not used during model fitting. The primary representation was a six-dimensional biophysical vector

$$\theta_v = (\Delta H_v,\ \Delta V_v,\ \Delta Q_v,\ f_v,\ s_v,\ h_v)\,, \tag{1}$$

where the first three components describe mutant-minus-wild-type differences in Kyte–Doolittle hy-

drophobicity, side-chain volume and side-chain charge at pH 7.0 (+1 for arginine and lysine, −1 for aspartate and glutamate and +0.1 for histidine). For a residue property $p(\cdot)$ and mutant and wild-type amino acids a_v^{mut} and a_v^{wt},

$$\Delta p_v = p(a_v^{\mathrm{mut}}) - p(a_v^{\mathrm{wt}}). \tag{2}$$

These differences were standardized before model fitting within each generalization setting, using the mean and standard deviation of that setting's training variants alone, so held-out variants do not influence the feature scale. The three position annotations below are on a common 0–1 scale and were not rescaled.

The remaining components encode position-level prior annotations, assigned according to the prespecified gene-specific rules in Supplementary Table 1. The fold-core term f_v was set to 1.0 for residues within an annotated structural core and 0.0 otherwise; the functional-switch term s_v to 1.0 for the listed switch residues (KRAS G12, G13 and Q61 and the zinc-coordinating cysteines of GATA1); and the hotspot or pathogenic marker h_v to 1.0 from external annotations alone, without reference to the single-cell response, using COSMIC and IARC missense hotspot codons, ClinVar pathogenic annotations and gene-specific functional-module membership.

The physicochemical difference terms are naturally defined for single amino-acid substitutions. Conditions without a single wild-type-to-mutant substitution were assigned zero for these three terms while retaining their position-based annotations; conditions naming multiple substitutions used the mean of the corresponding site-level differences. Because this representation is not injective, distinct variants can share the same feature vector and therefore receive identical predictions from a feature-based model. Such ties are handled explicitly by the tie-aware discrimination score defined below; sensitivity to alternative molecular representations is reported in Supplementary Note 2.

Protein-language-model embeddings. As an alternative molecular representation, per-variant embeddings were extracted from ESM-1v (`esm1v_t33_650M_UR90S_1`)[32]. For the mutant sequence of variant v, let $\mathbf{H}_v \in \mathbb{R}^{L_v \times 1280}$ denote the final-layer hidden states over L_v residues. The sequence embedding was obtained by mean pooling across residue positions,

$$e_v = \frac{1}{L_v} \sum_{\ell=1}^{L_v} \mathbf{H}_v[\ell, :] \in \mathbb{R}^{1280}, \tag{3}$$

and was used directly without dimensionality reduction. Synonymous variants were represented by the corresponding wild-type protein sequence. Three JAK1 conditions for which a well-formed mutant protein sequence could not be constructed were excluded from ESM-based analyses, leaving 467 of the 470 benchmark variants embedded. Mutant sequences longer than 1,000 residues were reduced to a 1,000-residue window starting 500 residues before the first listed substitution, clamped to lie inside the sequence, and every sequence was then truncated to its first 1,022 residues to fit the model's 1,024-token limit; wild-type reference sequences were embedded without windowing. Only JAK1 (1,154 residues) exceeds either limit, so its conditions are embedded over windows that differ between conditions and JAK1 ESM-1v features carry a condition-dependent crop. Matched-window mutation representations were evaluated separately with ESM2 (Supplementary Note 2).

Allele-specific prediction

Prediction target. For variant v, let $\mathcal{X}_v$ and $\mathcal{X}_{\mathrm{ref}}$ denote the corresponding variant and reference cells, and let $x_i \in \mathbb{R}^G$ denote the standardized expression profile of cell i over G genes. The measured perturbation response was defined as the pseudobulk difference

$$\delta_v = \frac{1}{|\mathcal{X}_v|} \sum_{i \in \mathcal{X}_v} x_i - \frac{1}{|\mathcal{X}_{\mathrm{ref}}|} \sum_{j \in \mathcal{X}_{\mathrm{ref}}} x_j. \tag{4}$$

For each profile, up to 300 cells were sampled without replacement from the variant and reference groups; variants represented by fewer than five cells were excluded. Each predictor maps molecular features of a held-out variant to an estimated response $\hat{\delta}_v$.

Generalization settings. We evaluated four primary forms of within-gene generalization. In each setting, models were fitted only to the training variants of a gene and evaluated on its held-out variants.

1. **Random holdout.** We used the predefined per-dataset partitions distributed with the allele-resolved component of PertResolve-Bench, with GATA1 following the PerturbNet partition. The held-out sets contained 16 of 98 TP53 variants, 8 of 92 KRAS variants, 52 of 254 GATA1 variants and 9 of 26 JAK1 variants.
2. **Positional extrapolation.** Variants located in the C-terminal half of each protein were held out, testing generalization to protein positions not represented by the corresponding training partition.
3. **Mechanistic extrapolation.** Variants at externally annotated hotspot or functional-switch residues were held out, testing generalization to functionally prioritized positions.
4. **Low-depth evaluation.** Variants represented by fewer than 200 cells were held out to evaluate prediction in the shallow-sampling regime. This setting applies to GATA1 (45 variants) and JAK1 (17 variants); all TP53 and KRAS variants exceed this depth.

A fifth *compatibility* setting reproduces the PerturbNet evaluation protocol. It uses the same 52 held-out GATA1 variants as the random setting together with predefined TP53 and KRAS holdouts. All variants evaluated under this setting are contained within the random-holdout evaluation.

A sixth *cross-gene transfer* setting, distributed with the benchmark, holds out GATA1 and JAK1 as whole genes (Fig. 1g); it is not scored, because the four datasets are measured in different cell types over different gene panels and share no common expression space.

Feature-based predictors. The primary model grid comprised 18 feature–model combinations obtained by crossing six regression heads with three molecular feature spaces. Each model learns

$$g_\phi : \mathbb{R}^p \to \mathbb{R}^G \tag{5}$$

from a p-dimensional variant representation to the G-dimensional perturbation response and was fitted on training variants by minimizing

$$\min_\phi \sum_{v \in \text{train}} \left\| g_\phi(\text{feat}_v) - \delta_v \right\|_2^2, \tag{6}$$

with model-specific regularization where applicable.

The six regression heads were ridge regression, lasso regression, random forest, gradient-boosted trees, k-nearest-neighbours regression and a multilayer perceptron. Hyperparameters were set to the values listed in Supplementary Table 2 for every gene, generalization setting and split, without hyperparameter search. Linear, neighbour and neural-network models predicted the expression profile directly. For expression spaces exceeding 200 genes, random-forest and gradient-boosted models were fitted to a principal-component representation of the training responses and transformed back to gene space after prediction. Principal components were fitted using training responses only.

The three primary feature spaces were the six-dimensional biophysical representation θ, the 1,280-dimensional ESM-1v embedding, and their concatenation. Additional representation controls included ESM2-based mutant embeddings, mutant-minus-wild-type ESM2 representations at global, mutated-residue and local-window scales, and AlphaMissense pathogenicity scores. These controls used the same train–test partitions and evaluation metrics; representation-specific construction details are given in Supplementary Note 2.

Reference predictors. The WT-null reference predicts

$$\hat{\delta}_v = 0 \tag{7}$$

for every variant and represents prediction without perturbation information. The Gene-mean reference predicts the mean response of the training variants of the held-out variant's gene,

$$\hat{\delta}_v = \frac{1}{|\text{train}(g_v)|} \sum_{w \in \text{train}(g_v)} \delta_w. \tag{8}$$

Because this reference uses gene identity but no allele-specific molecular information, it quantifies the response recoverable from gene-shared structure.

Released-model evaluation. Released perturbation-prediction frameworks were evaluated with the same pseudobulk targets, held-out variants and PDS definition as the feature-based predictors (Supplementary Table 3; implementation details in Supplementary Note 3). Biolord[33] and CellFlow[7] were run using their released implementations, with variant ESM-1v embeddings as the CellFlow condition. Because scGen[4] indexes categorical perturbations, we evaluated an allele-conditioned adaptation in which its latent shift is a fitted function of molecular variant attributes and is therefore defined for a variant never seen in training. A corresponding scVIDR[34] adaptation additionally scales this shift by a per-variant variable, an external AlphaMissense pathogenicity score for TP53 and KRAS[30] but a quantity derived from measured response cells for GATA1; scVIDR is therefore reported as a response-informed control and enters no response-blind comparison.

scGPT[8], GEARS[6] and STATE[35] condition on perturbed-gene identity and produce no within-gene allele distinction. PerturbNet[31] was evaluated with ESM-1v conditioning across all five scored within-gene settings (492 gene–setting–variant evaluations, 303 distinct variants) in its 50-dimensional principal-component output representation, with scores aggregated as for the primary PDS estimator (Statistical analysis).

Prediction evaluation

PertResolve-Eval assesses complementary properties of perturbation prediction. For the allele-resolved benchmark, the primary metric, perturbation discrimination score (PDS), measures whether a prediction is specific to the correct allele. Direction-recovery metrics quantify agreement with the overall transcriptional response, and differential-expression metrics assess agreement in the genes and effect sizes underlying that response. Metrics are calculated for each held-out variant and aggregated over distinct held-out variants (Statistical analysis).

Perturbation discrimination score. For a held-out variant v of gene g, the predicted response $\hat{\delta}_v$ was compared with the measured responses of all variants of the same gene. The candidate set

$$\mathcal{C}_g = \mathrm{train}(g) \cup \mathrm{test}(g) \tag{9}$$

therefore contains both training and held-out variants that pass the five-cell filter. Let $n = |\mathcal{C}_g|$, let

$$\ell_v = \left|\left\{w \in \mathcal{C}_g : d(\hat{\delta}_v, \delta_w) < d(\hat{\delta}_v, \delta_v)\right\}\right| \tag{10}$$

be the number of candidates strictly closer to the prediction than its true target, and let

$$e_v = \left|\left\{w \in \mathcal{C}_g : d(\hat{\delta}_v, \delta_w) = d(\hat{\delta}_v, \delta_v)\right\}\right| \tag{11}$$

be the number tied with the true target, including v itself. We define the tie-aware score

$$\mathrm{PDS}(v) = 1 - \frac{\ell_v + \frac{1}{2}(e_v - 1)}{n - 1}. \tag{12}$$

The nearest unique match scores 1, whereas random matching has expected score 0.5; ties receive their mid-rank. Only held-out variants are scored, but each prediction competes against the full same-gene candidate set. Predictions yielding no finite ranking, including zero-norm or non-finite vectors, and candidate sets containing fewer than two variants are assigned a score of 0.5.

Top-k shortlisting. The same ranking supports a tail statistic. Using ℓ_v and e_v above, the probability that the true allele falls in the top k of its same-gene ranking when ties are broken at random is

$$\mathrm{Hit}_k(v) = \frac{\mathrm{clip}(k - \ell_v,\, 0,\, e_v)}{e_v}, \tag{13}$$

evaluated at $k \in \{1, 3, 5, 10\}$, the same shortlist lengths used for the measurement reference. The chance rate is computed per query from that query's own candidate-pool size, $\min(k/N_v, 1)$, because pool size differs by gene (98, 92, 254 and 26). A prediction with no finite distance to its target has no ranking and is assigned its query's chance expectation, matching the convention by which such predictions receive

PDS 0.5. Seed, split and query aggregation and the bootstrap follow the PDS estimator exactly, and one bootstrap draw scores all four k, so the recovery curve retains its covariance. Inference is the paired bootstrap of the observed rate minus that method's own chance rate over the same resampled queries.

Cosine distance was used for the primary analysis,

$$d_{\cos}(a,b) = 1 - \frac{a^\top b}{\|a\|_2 \|b\|_2}. \tag{14}$$

Robustness analyses additionally used

$$d_1(a,b) = \|a-b\|_1, \qquad d_2(a,b) = \|a-b\|_2. \tag{15}$$

The null references for PDS are given in Statistical analysis.

Direction recovery and residualization. Agreement with the measured transcriptional response was quantified using Pearson correlation,

$$\text{Pearson-}\delta = \frac{1}{n_{\text{test}}} \sum_{v \in T} \operatorname{corr}\left(\hat{\delta}_v, \delta_v\right), \tag{16}$$

where T denotes the held-out variants.

The descriptive residual Pearson analysis in Fig. 2e (left) centred measured and predicted profiles separately across the same evaluated variants of each gene. Each distinct variant contributed its first available prediction in the fixed evaluation-setting order. Writing T_g for that cohort, the profiles entering the correlation were $\delta_v - |T_g|^{-1} \sum_{w \in T_g} \delta_w$ and $\hat{\delta}_v - |T_g|^{-1} \sum_{w \in T_g} \hat{\delta}_w$. Correlations were averaged across variants with non-constant residual profiles.

For residual PDS in Fig. 2e (right), we instead subtracted the same training-only gene mean from both measured and predicted profiles,

$$\bar{\delta}_g = \frac{1}{|\text{train}(g)|} \sum_{w \in \text{train}(g)} \delta_w, \tag{17}$$

giving

$$r_v = \delta_v - \bar{\delta}_g, \qquad \hat{r}_v = \hat{\delta}_v - \bar{\delta}_g. \tag{18}$$

PDS was then recomputed on these residual profiles and compared with its own task-matched permutation distribution (Statistical analysis).

Measured-response decomposition. The measurement-only decomposition in Fig. 2d used all eligible benchmark variants of each gene, independently of model-training assignments. Variants required at least five cells; the eligible sets $\mathcal{V}_g$ contained 98 TP53, 92 KRAS, 254 GATA1 and 26 JAK1 variants. For each of $B = 15$ random cell partitions, variant and reference cells were divided into disjoint halves, each capped at 300 cells. Let $\delta_v^{(b)}$ denote the variant-minus-reference mean response from the second half of partition b. The gene-shared mean, mean squared response norm and mean squared residual norm were

$$\begin{aligned} \mu_g^{(b)} &= \frac{1}{|\mathcal{V}_g|} \sum_{v \in \mathcal{V}_g} \delta_v^{(b)}, \\ T_g^{(b)} &= \frac{1}{|\mathcal{V}_g|} \sum_{v \in \mathcal{V}_g} \|\delta_v^{(b)}\|_2^2, \\ A_g^{(b)} &= \frac{1}{|\mathcal{V}_g|} \sum_{v \in \mathcal{V}_g} \|\delta_v^{(b)} - \mu_g^{(b)}\|_2^2. \end{aligned} \tag{19}$$

The displayed allele-residual percentage was

$$100 f_g, \qquad f_g = 1 - \frac{1}{B} \sum_{b=1}^{B} \frac{\|\mu_g^{(b)}\|_2^2}{T_g^{(b)} + \epsilon}, \qquad \epsilon = 10^{-12}. \tag{20}$$

Because residuals sum to zero within each partition, $T_g^{(b)} = \|\mu_g^{(b)}\|_2^2 + A_g^{(b)}$. Thus, apart from the numerical stabilizer, f_g is the mean of the partition-specific squared-residual fractions $A_g^{(b)}/T_g^{(b)}$, with equal weight per variant within each partition. The complementary gene-shared percentage is $100(1 - f_g)$.

Differential-expression fidelity. For each variant, expression genes were ranked using a Welch t-test comparing variant and reference cells, with at most 300 cells per group. Let $\mathcal{D}_v^{\text{true}}$ denote the $k = 50$ genes with the largest absolute t-statistics and $\mathcal{D}_v^{\text{pred}}$ the k genes with the largest absolute predicted responses. Their overlap was

$$\text{DE overlap}(v) = \frac{|\mathcal{D}_v^{\text{pred}} \cap \mathcal{D}_v^{\text{true}}|}{k}. \tag{21}$$

The measured effect size of gene g in variant v was the difference of group means on the same standardized expression scale used for the t-test,

$$\lambda_{v,g} = \frac{1}{|S_v|} \sum_{i \in S_v} x_{i,g} - \frac{1}{|S_{\text{ref}}|} \sum_{j \in S_{\text{ref}}} x_{j,g}, \tag{22}$$

where S_v and S_{ref} are the variant and reference draws entering the test. $\lambda_{v,g}$ is a standardized mean difference in units of the gene's standard deviation. We call the metric *DE effect-size Spearman*. It is Spearman's rank correlation between $\lambda_{v,g}$ and $\hat{\delta}_{v,g}$ over $\mathcal{D}_v^{\text{true}}$, with ties receiving mid-ranks. Direction agreement is the fraction of those genes for which $\text{sign}(\hat{\delta}_{v,g}) = \text{sign}(\lambda_{v,g})$. The cell draws entering the differential-expression statistics are made independently of those forming the pseudobulk response used for PDS and Pearson-δ, although both take the same reference population.

Measurement resolution

Perturbation detection. We first asked whether each perturbation produced a response that was reproducibly distinguishable from its reference at the available sampling depth. For two cell sets $X = \{x_i\}_{i=1}^n$ and $Y = \{y_j\}_{j=1}^m$, the energy distance was

$$E(X, Y) = \frac{2}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} \|x_i - y_j\|_2 - \frac{1}{n^2} \sum_{i,i'} \|x_i - x_{i'}\|_2 - \frac{1}{m^2} \sum_{j,j'} \|y_j - y_{j'}\|_2. \tag{23}$$

For each perturbation with sufficient cells, cells were randomly divided into two matched halves A and B, and a size-matched sample C was drawn from the corresponding reference population. We calculated the within-perturbation split-half distance

$$D_{\text{self}} = E(A, B) \tag{24}$$

and the perturbation-to-reference distance

$$D_{\text{null}} = E(A, C). \tag{25}$$

The analysis was repeated over 50 random cell splits. The primary analysis used a 50-component principal-component representation at the deepest split-half depth supported by each perturbation. The principal-component basis was fitted once per gene for the allele-resolved datasets and once per dataset for the gene-perturbation screens, on all of that gene's or dataset's cells, perturbed and reference alike. The allele-resolved datasets entered in their native expression spaces; the three gene-perturbation screens were library-size normalized to 10^4 counts per cell, log transformed and restricted to their 2,000 most highly variable genes before the basis was fitted. It was fitted before any cell split and was not refitted within a depth rung or a split. The basis is unsupervised and shared by every cell set entering D_{self} and D_{null}. The corresponding basis for the broader perturbation panel is described with that criterion below.

Detection eligibility was defined by comparing the separation from the reference with the sampling variation of the perturbation itself. Let

$$S = \text{med}(D_{\text{null}}) - \text{med}(D_{\text{self}}) \tag{26}$$

and

$$W = Q_{97.5}(D_{\text{self}}) - Q_{2.5}(D_{\text{self}}). \tag{27}$$

A perturbation was classified as detection-eligible when

$$S > W. \tag{28}$$

This definition was applied identically to the four allele-resolved datasets and to the Replogle, Norman and Adamson gene-perturbation datasets obtained through scPerturb[13,14,36,37]. At native depth, each perturbation was evaluated at the deepest rung of the split-half subsample ladder that it supported: 30, 50, 100 and 150 cells per half for the four allele-resolved datasets, and 25, 50, 100, 200, 400 and 800 for the gene-perturbation datasets. A perturbation entered a rung only when it supplied at least twice that rung's cells, which admitted 464 of the 470 allele-resolved variant conditions; the six excluded are JAK1 variants holding fewer than 60 cells. Allele-resolved analysis pools were capped at 300 cells per perturbation, so native depth denotes the deepest evaluated subsample rather than every available cell; 341 of the 464 perturbations were evaluated at the deepest rung of 150 cells per half. Alternative distance functions, detection criteria, expression representations and reference variability were examined in Supplementary Note 1.

Sibling identification. Identification among sibling perturbations was quantified using pseudobulk response profiles constructed from disjoint cell samples. For variant v, let δ_v^A and δ_v^B denote response profiles constructed from disjoint cell halves at matched depth. Within-variant sampling variation was measured by

$$D_{\text{self}}(v) = d_{\cos}\left(\delta_v^A, \delta_v^B\right). \tag{29}$$

For two sibling variants v_i and v_j, their between-variant separation was measured in the same response space. A pair was classified as resolvable when its between-variant distance exceeded the average split-half distance of the two variants,

$$d_{\cos}\left(\delta_{v_i}, \delta_{v_j}\right) > \frac{D_{\text{self}}(v_i) + D_{\text{self}}(v_j)}{2}. \tag{30}$$

At the variant level, identification used the more stringent nearest-sibling criterion. Both statistics were averaged over the 15 split-half repeats before any threshold was applied, giving $\bar{D}_{\text{self}}(v)$ and, for the between-variant distances, $\bar{D}(v_i, v_j)$ computed between the two variants' A-half profiles. Writing $C(v_i)$ for the other scored variants of the same gene, variant v_i was classified as identifiable when

$$\min_{v_j \in C(v_i)} \bar{D}(v_i, v_j) > \bar{D}_{\text{self}}(v_i), \tag{31}$$

with exact ties classified as not identifiable. This criterion selects the nearest sibling on the same seed-averaged distance matrix and is distinct from the cross-fitted nearest-competitor statistic defined below.

Split-half reproducibility reference. To quantify the full-pool identification supported by the measurement itself, we constructed a split-half reproducibility reference using the same PDS evaluation applied to prediction models. For each held-out variant, its cells and the corresponding reference cells were divided into two disjoint groups. Pseudobulk responses were constructed separately from the two disjoint halves, and one response was treated as the prediction while the other was treated as the evaluation target. PDS was then calculated using the same candidate variants, distance function, tie handling and generalization setting as in the model evaluation.

The procedure was repeated over 15 random split-half subsamples and averaged for each variant before aggregation. This quantity measures within-experiment cell-sampling reproducibility at the specified sampling depth and protocol. It is an empirical reference rather than a mathematical bound: predictors that denoise responses or borrow information across variants can in principle exceed it.

The same ranking also supports a shortlist reading. For each held-out variant and each split-half repeat we retained the rank of the true allele among the same-gene candidates instead of only its percentile, and recorded whether that rank fell within $k \in \{1, 3, 5, 10\}$. Ties were resolved as the expected hit

under uniform tie-breaking, the indicator analogue of the mid-rank convention PDS uses. Hit rates were averaged over the 15 repeats and then over distinct variants, with 2,000-resample percentile bootstraps over distinct variants. Because every split's candidate set is that gene's full scored pool, a variant held out under several generalization settings receives the same score under each, and the effective sample is the 308 distinct variants. The chance rate is $\min(k/N, 1)$, with N equal to 98, 92, 254 and 26 for TP53, KRAS, GATA1 and JAK1.

The split-half reference draws both groups from a pooled shuffle, so both carry the same batch composition. For datasets with per-cell experimental batch information, the same reference was recomputed with the two groups drawn from disjoint batches (Supplementary Note 1).

Pairwise sibling discrimination. Full-pool identification requires a prediction to outrank an entire candidate set. Pairwise discrimination asks the narrower question of whether a prediction separates two siblings: the candidate set is two by construction and chance is 0.5 irrespective of the gene's pool size.

For a pair of variants v_i and v_j, a predictor resolves the pair when the correct assignment of its two predictions to the two measured responses is closer than the swap,

$$d(\hat{\delta}_i, \delta_i) + d(\hat{\delta}_j, \delta_j) < d(\hat{\delta}_i, \delta_j) + d(\hat{\delta}_j, \delta_i), \tag{32}$$

with d the cosine distance used throughout; exact ties score 0.5. The measurement arm applies the identical statistic with one disjoint cell half in place of the prediction, so both arms are scored against the same half-depth target.

A pair was called measurement-resolvable under the criterion already used for allele identification, namely a between-variant distance exceeding the mean of the two variants' split-half distances. Pair selection and pair evaluation used disjoint blocks of subsample seeds. Only held-out variants with model predictions were scored, and a pair held out under several generalization settings was averaged across them before entering the bootstrap, so that no pair contributed more than once. The displayed confidence intervals are 2,000-resample percentile bootstraps over distinct pairs; because distinct pairs can share a variant, a variant-node bootstrap that applies the same draw to the measurement and every model is reported in Supplementary Note 2. Predictions containing non-finite values were excluded from pairwise scoring and their counts are reported alongside the accuracies.

Nearest-competitor separation. Because PDS is a ranking statistic, we quantified each perturbation's separation from its nearest competitor relative to within-condition sampling noise. At depth m, each variant contributes two profiles from disjoint groups of m cells, and the per-profile noise $\hat{\eta}^2$ is their pooled within-variant variance. Mean pairwise separation ρ^2 is the noise-corrected mean squared distance between variant means, divided by $2\hat{\eta}^2$. Nearest-competitor separation was estimated by cross-fitting: the nearest sibling was selected on one profile and its squared distance read on the other, symmetrized over the two directions and corrected by $2\hat{\eta}^2$. This prevents the same sampling fluctuation from both selecting a competitor and making it appear close. Separations were computed within each of 10 cell partitions, averaged across partitions and summarized as ρ^2_{nn}, their median over variants divided by $2\hat{\eta}^2$. Under a working model of independent, mean-zero and exchangeable per-profile sampling errors with a common variance, zero is the nominal no-signal reference for both statistics; with variant-specific variances the pooled correction can shift this centring. Estimates are reported unclipped. Intervals are 95% percentile bootstraps of the median over variants, with 2,000 resamples, holding $\hat{\eta}^2$ and the cell partitions fixed.

To distinguish effects of average signal magnitude from response geometry, we additionally generated controlled synthetic response configurations in which separation rank, amplitude distribution, candidate-set size and sampling depth were varied while overall signal relative to within-condition variation was controlled. Each configuration was passed through the same split-half PDS procedure. Mean pairwise separation and cross-fitted nearest-competitor separation were then compared with the attainable reproducibility reference across measured and controlled-resolution configurations. Simulation construction and the normalized separation metrics are described in Supplementary Note 4.

Benchmark-design analyses

Controlled predictor family. To measure whether a benchmark can reliably distinguish predictors, we constructed a family of predictors whose relative quality was fixed before evaluation. For variant v,

$$\hat{\delta}(v, \alpha) = (1 - \alpha)\bar{\delta}_g + \alpha\delta_v^{\text{build}}, \tag{33}$$

where $\bar{\delta}_g$ is the gene-shared mean response and δ_v^{build} is a variant-specific response estimated from cells disjoint from those used for evaluation. Increasing α increases the contribution of variant-specific information and therefore defines an ordered predictor family independently of the evaluation data. The weights were $\alpha \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and the depths $m \in \{25, 50, 100, 150, 250\}$ cells per half. For each gene, depth and draw, a variant's cells were shuffled once and cut into a build half and a disjoint evaluation half of m cells, and the reference cells were cut by the same shuffle into two disjoint halves, so a build profile and its evaluation target share neither perturbed cells nor a reference draw. A variant entered a depth only if it held at least $2m$ cells, and only cohorts identical across draws were used. The gene-shared anchor is the mean of the build profiles over that gene-by-depth cohort, so both anchors are estimated in build cells alone. Because every variant receives the same prediction at $\alpha = 0$, the cohort-mean PDS there is exactly 0.5 by construction rather than an estimate.

Each predictor was scored by PDS against the held-out evaluation cells. Bootstrap resampling of held-out perturbations was then used to estimate two benchmark-level quantities: the probability of recovering the complete design-defined ordering of the predictor family and the probability of selecting the highest-quality predictor. These quantities were evaluated as functions of the split-half reproducibility reference and the number of perturbations available for scoring.

Each predictor's per-variant PDS was averaged over 10 subsample draws before aggregation, and the ordering and winner probabilities were estimated from 1,000 bootstrap resamples of the cohort's variants.

For the six screens in Supplementary Table 5, the same predictor construction was evaluated within the broader-panel criterion described below, over the seven weights $\alpha \in \{0, 0.5, 0.7, 0.85, 0.925, 0.96, 1.0\}$, whose spacing narrows towards $\alpha = 1$, averaging the eight cell partitions and bootstrapping 1,000 times. These values are compared within the panel rather than against the allele-resolved numbers.

Candidate-set composition. Candidate-set size and candidate-set composition were varied separately, recomputing the split-half reproducibility reference at each candidate-set definition with the same PDS evaluation used for model scoring. Here K is the total candidate-set size, including the target. In the uniform arm, $K - 1$ competitors were drawn without replacement from the same-gene pool, always retaining the true target, at $K \in \{5, 10, 20, 25, 45, 90\}$ and the full pool, with 200 draws per seed. In the nearest-competitor arm, the $K - 1$ competitors closest to the target were retained, selected on the cell half disjoint from the one used for scoring. For comparison across genes, we also evaluated the common subset size $K = 25$; the complete JAK1 pool contained $K = 26$ candidates.

Uniform subsampling leaves the expected PDS unchanged, so it changes the resolution of the percentile but not its expectation (Supplementary Note 4); counts were sampled from the corresponding multivariate hypergeometric distribution.

Sampling-depth analysis. To quantify how detection changes with measurement depth while holding perturbation identity fixed, the split-half analysis was repeated at 30, 50, 100 and 150 cells per half. A perturbation entered a depth condition only when at least twice the required number of cells was available. Within each gene, comparisons across depths used the cohort of perturbations evaluable at the deepest depth being compared, so changes along a depth curve reflect sampling depth rather than changing perturbation composition.

For the broader gene-perturbation datasets, native-depth detection was additionally compared with detection after subsampling to 50 cells per half using only perturbations evaluable under both conditions. The same $S > W$ criterion was used throughout.

Repeated-pilot aggregate for detection association. For each perturbation in Replogle, Norman and Adamson, perturbed cells were randomly permuted ten times; in each permutation the first 50 cells formed the pilot pool and the next 200 cells two 100-cell evaluation halves, so pilot and evaluation cells were disjoint within a permutation but could recur across permutations. Cells were represented in 50 components fitted once per screen before partitioning, so evaluation cells contributed to this unsupervised basis (Supplementary Table 2). From the first $P \in \{25, 50\}$ pilot cells we computed a pilot effect magnitude and a training-free pilot signal-to-variation score,

$$\mathrm{eff}_P = \left\|\bar{x}_{\mathrm{pilot},P} - \bar{x}_{\mathrm{ref},P}\right\|_2, \qquad \mathrm{SNR}_P = \frac{\mathrm{eff}_P}{\left\|\bar{x}_{1:h} - \bar{x}_{h+1:2h}\right\|_2}, \qquad h = \lfloor P/2 \rfloor, \tag{34}$$

where $\bar{x}_{\mathrm{ref},P}$ is the centroid of P reference cells and the denominator of SNR_P is the distance between the centroids of two disjoint halves of the pilot pool, so that it measures within-perturbation cell-sampling variation alone. Both scores were averaged over the ten permutations. The detection label at 100 cells per half was $\mathbf{1}(S > W)$ computed from the ten evaluation draws, with S the median evaluation perturbation-to-reference distance minus the median split-half distance and W the range between the 2.5th and 97.5th percentiles of the ten split-half distances.

The learned score was a leave-one-dataset-out logistic regression on standardized eff_P; with one feature and a positive coefficient it is a monotone function of eff_P, so the fit affects calibration rather than ranking. The training-free score used SNR_P directly. AUROC was calculated within each held-out dataset rather than after pooling, and not for the allele-resolved datasets, because TP53 and KRAS carry a single eligibility class. Intervals are 95% percentile bootstraps over perturbations with 2,000 resamples, discarding resamples lacking either class; tied scores contribute one half.

Sampling-depth projection. The controlled-geometry sweep was inverted into a sampling-depth projection, given in full in Supplementary Note 5. The axis is the median cross-fitted nearest-competitor separation ρ^2_{nn} and the calibrated outcome is the split-half reproducibility reference, each selected on the sweep. Within three candidate-set-size strata, the outcome was calibrated against the axis by binned medians made monotone by a running maximum, with no assumed functional form. The nearest-competitor separation is taken to be independent of sampling depth, whereas the per-profile noise scales as $\hat{\eta}^2 \propto 1/m$, so the depth projected from a pilot at m_0 cells per group is

$$\hat{m}(\tau) = m_0 \, \frac{x^*(n,\tau)}{\rho^2_{\mathrm{nn}}(m_0)}, \tag{35}$$

where $x^*(n,\tau)$ is the smallest calibrated axis value reaching the target $\tau \in \{0.7, 0.9\}$ on the reproducibility reference at candidate-set size n. No depth is returned when the calibration does not reach τ, when the pilot estimate is not positive or when its lower confidence bound does not exceed zero; the last condition prevents separations whose intervals cover zero from returning depths that would be extrapolations from noise. The calibration includes the twelve measured gene-by-depth configurations, so projections for those datasets are in-sample with respect to it.

Benchmark-size resolution. The number of variants required to compare models was measured on the design-ordered predictor family. Difficulty was indexed by the realized full-cohort PDS difference between two predictors rather than by their weight difference, because the map from weight to score saturates; all 15 weight pairs were used. For each gene and depth, held-out variants were subsampled *without* replacement at $N_{\mathrm{test}} \in \{5, 10, 20, 40, 80, 160\}$, 200 draws each, and the reported quantity is the fraction of draws in which the better predictor received the higher mean score; full-cohort evaluations are deterministic and are excluded from the curves.

Conditions were grouped by the split-half reproducibility reference, taken as the score of the $\alpha = 1$ member on the evaluation half ($r \leq 0.52$, $0.52 < r \leq 0.65$ and $r > 0.65$, with 0.65 the high-reproducibility operating point of the broader panel), and by the realized difference ($\Delta \leq 0.01$, $0.01 < \Delta \leq 0.05$ and $\Delta > 0.05$). Both sets of edges were fixed before this analysis. Only weight pairs with a positive realized difference entered, so the probabilities are conditional on the design ordering being realized: of the 945 gene–depth–pair–size conditions, 265 had a non-positive difference, all in TP53 and KRAS, and 680 entered the curves. Within a bin, recovery probabilities were pooled across the contributing conditions at each N_{test}, a set that can change across sizes, and we report the smallest tested N_{test} at which the mean reaches 0.9, written ≤ 5 when it is the smallest size tested; no monotonicity was imposed.

Broader perturbation panel

Prespecified criterion.

The broader arm of PertResolve-Bench evaluates measurement resolution beyond allele-resolved perturbation data using a prespecified three-part criterion for perturbation detection, perturbation iden-

tification and benchmark-level ranking resolution. Eligible screens required a defined perturbation label, an identifiable reference condition and at least five perturbations represented by at least 200 cells.

The criterion and decision thresholds were specified on six screens: Norman, Replogle, Adamson, the Virtual Cell Challenge training dataset[38], the multiplexed genetic and chemical perturbation screen of McFarland *et al.*[39], and a subset of Tahoe-100M[40]. They were then applied unchanged to an expanded panel comprising 31 configurations across 14 public resources[13,14,36,38–47]; Parse 10M is documented as a public dataset release in Supplementary Data 1. A configuration denotes one resource evaluated within one cellular context and one readout modality; configurations are therefore nested within resources rather than statistically independent resources. The panel represents a selected evaluation set under the stated eligibility criteria rather than an exhaustive census of eligible public datasets. McFarland and Tahoe-100M were evaluated using predefined subsamples of their larger resources.

For each configuration, the primary analysis used 50 cells per group, eight random cell partitions, 1,000 bootstrap resamples and a 50-component principal-component representation of the stored RNA count matrix, fitted separately within each resource and context before cell partitioning. For readouts with at most 50 measured features, including the surface-protein datasets, distances were computed directly in the measured feature space. A screen was classified as detectable when more than 50% of perturbations met the perturbation-versus-reference criterion and identifiable when more than 50% met the nearest-competitor criterion; a split-half reproducibility reference above 0.65 defined the high-reproducibility operating point.

Count-space panel pipeline and depth-ladder pipeline. Two pipelines produce the detection fractions reported in this work. The *panel criterion* computes distances in 50 principal components of the unnormalized count matrix, with no library-size normalization, no log transformation and no gene selection, over eight random cell partitions, including a perturbation when it holds at least 200 cells. It produces every detection fraction in Fig. 5 and the reference arm of Supplementary Fig. 2. The *split-half depth ladder* computes distances in 50 principal components of each dataset's own analysis space, which for the three gene-perturbation screens is library-size-normalized, log-transformed expression restricted to 2,000 highly variable genes and for the allele-resolved datasets is the native space of that dataset, over 50 random cell splits, including a perturbation when it holds at least twice the depth being tested; it is the pipeline behind Fig. 4c,d and Fig. 6f. The two pipelines and the two cell-inclusion thresholds each shift the absolute detection fraction, in opposite directions; on the three resources evaluated under both pipelines, the representation alone shifted it by 0.097–0.195 (Supplementary Fig. 2). Detection fractions are therefore compared only within a pipeline and inclusion threshold.

Statistical analysis

The primary statistical unit for model evaluation was the distinct protein-coding variant. The primary comparison contains 518 held-out evaluations corresponding to 308 distinct variants, or 511 evaluations and 305 distinct variants for predictors requiring a sequence embedding, because three JAK1 conditions have no constructible mutant sequence. For PDS, scores were first averaged over 15 random cell subsamples within each gene–setting–variant unit. When a variant was evaluated under more than one generalization setting, these setting-level means were then averaged before aggregation over distinct $(\text{gene}, \text{variant})$ pairs. Point estimates and 95% confidence intervals were obtained using a 2,000-resample nonparametric percentile bootstrap over these pairs. The differential-expression summaries in Fig. 2f instead average each predictor's values over all of its held-out evaluations. Confidence intervals excluding 0.50 were interpreted against the exact full-candidate uniform-matching expectation.

Pearson-δ used a fixed pseudobulk draw and was aggregated over the same distinct-variant unit, averaging across settings first. Non-finite prediction vectors were excluded from Pearson-δ rather than represented by zero, and their coverage is reported separately. Point estimates and 95% confidence intervals were calculated using the same 2,000-resample percentile bootstrap over distinct $(\text{gene}, \text{variant})$ pairs. Positive direction recovery was defined by a lower 95% confidence bound above zero.

The tie-aware mid-rank definition gives an exact expectation of 0.50 when the target label is assigned uniformly over every column of the complete candidate pool. For the primary feature-based predictions, permuting prediction rows 1,000 times within held-out-only distance matrices for each gene and evaluation setting gave mean PDS values of 0.494–0.505 across 1,020 predictor, gene, setting and distance

combinations (grey band in Fig. 2c), consistent with this expectation.

Task-matched permutations for the residual analysis and PerturbNet instead retained all training and held-out candidate columns used by the reported PDS while reassigning prediction rows only among the available held-out target labels within each gene and evaluation setting. The restricted held-out label set is not generally uniform over the complete ranking pool, so this conditional permutation distribution need not be centred at 0.50. The setting-specific distance matrix and, for PerturbNet, its 50-dimensional output representation were held fixed. PerturbNet used the same label permutation across its 15 cell-subsample seeds; scores were averaged first over seeds within gene–setting–variant, next across settings within gene–variant, and finally over 303 distinct variants, matching the reported estimator. With 1,000,000 permutations and random seed 0, its observed PDS was 0.530, the exact restricted-label null mean was 0.515, the central 95% permutation interval was 0.496–0.535 and the one-sided plus-one permutation value was 0.070. Residual-axis permutations used the same restricted-label/full-candidate principle. They were run in two stages, with 50 permutations for every method and 1,000 for methods whose preliminary value was below 0.05; residual-axis P values were adjusted across the 25 scored methods. Additional details are provided in Supplementary Note 2.

Detection and identification fractions were treated as binomial proportions and are reported with 95% Wilson score intervals. Native-versus-depth-matched detection comparisons used a paired 2,000-resample bootstrap over perturbations, with each resample applied to both depth conditions. Confidence intervals for continuous per-gene measurement quantities were obtained by nonparametric bootstrap over perturbations unless otherwise stated.

All stochastic analyses used fixed random seeds. No statistical method was used to predetermine the number of biological conditions. For the primary allele-resolved benchmark, all identified public datasets and variant conditions meeting the predefined eligibility criteria were retained. A residue-clustered bootstrap assessing dependence among substitutions at the same residue is reported in Supplementary Note 2.

Software

Primary analyses were performed in Python 3.11.14 using NumPy 2.2.6, pandas 2.3.3, SciPy 1.14.1 and scikit-learn 1.7.2. Single-cell data processing used scanpy 1.10.3 in Python and `scran` with `SingleCellExperiment` in R/Bioconductor. Protein-language-model embeddings and other GPU-accelerated analyses were performed using an NVIDIA GeForce RTX 4090 GPU. Embeddings were generated with fair-esm 2.0.0 and PyTorch 2.10.0 with CUDA 12.6.

Released-model implementations were run in package-specific environments. Biolord 0.0.3 and the scGen and scVIDR adaptations used scvi-tools 1.4.3, and CellFlow used `cellflow-tools` 0.1.1 and JAX.

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Author contributions

B.L., C.Z. and M.L. contributed to the conception and design of the study. B.L. led the methodology development, software implementation, data curation, computational analyses, visualization and manuscript preparation. C.Z. contributed to computational analyses, result interpretation and visualization. M.L. contributed to conceptual development, study design, result interpretation and manuscript revision. B.Z. provided supervision, research support and funding acquisition. L.W. contributed to conceptual discussions and manuscript revision. Z.T., J.Liu, C.L. and C.W. contributed to study design, result interpretation and manuscript revision. Y.Y. and J.Lv provided research supervision and analytical support. Y.Z. conceived and supervised the study, contributed to study design, analysis, interpretation and manuscript preparation, and provided project administration and funding acquisition. All authors

reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

All single-cell datasets analysed in this study are publicly available. TP53 and KRAS Perturb-seq data are available from the Gene Expression Omnibus under accession GSE161824[18]. GATA1 Perturb(BE)-seq data are available under accession GSE215253[19]; the processed expression matrix and predefined partitions used here were obtained from the PerturbNet Hugging Face repository (https://huggingface.co/cyclopeta/PerturbNet_reproduce)[31]. JAK1 scSNV-seq data are available from the European Nucleotide Archive under accession PRJEB48915, with processed objects deposited at Zenodo (10.5281/zenodo.10418435)[20]. The Replogle, Norman and Adamson datasets used in the measurement-resolution analyses were obtained through scPerturb[37]. Additional resources were obtained from the public releases documented in Supplementary Data 1, including KOLF2.1J[41], Papalexi[42], PerturbMulti[43], Frangieh[44], X-Atlas/Orion[45] and GSE306429[46]; the remaining source studies and accessions are listed there.

Derived PertResolve-Bench tables, including allele-resolved variant annotations, cell counts, evaluation partitions, per-variant prediction results and measurement-resolution results for the broader perturbation panel, are available at https://github.com/Boom5426/PertResolve. Expression matrices and protein-language-model embeddings, which are large or governed by source licences, can be regenerated from the public datasets with the preprocessing procedures described in Methods.

Code availability

Code for constructing PertResolve-Bench, fitting predictors, computing PertResolve-Eval metrics, performing measurement-resolution analyses and reproducing the main figures is available at https://github.com/Boom5426/PertResolve.

PertResolve-Eval is distributed as the `pertresolve` Python package with both Python API and command-line interfaces for applying the PertResolve workflow directly to new single-cell perturbation datasets, including AnnData (`.h5ad`) inputs. Given perturbation labels and a reference condition, the framework reports perturbation detectability, nearest-competitor identification, split-half reproducibility, local response geometry and benchmark-level model-ranking resolution.

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